

AMENDMENT NO. 4 AUGUST 2026
TO
IS 16444 (PART 1) : 2015 a.c. STATIC DIRECT CONNECTED WATTHOUR
SMART METER CLASS 1 AND 2 — SPECIFICATION

(Foreword, para 6) — Substitute the following for the existing:

‘The Electrotechnical department of Bureau of Indian Standards has prepared many metering standards such as IS 13779 ‘a.c. Static watthour meters, Class 1 and 2 — Specification’, IS 14697 ‘a.c. Static transformer operated watthour meters (Class 0.1 S, 0.2 S and 0.5 S) and var-hour meters (Class 0.1 S, 0.2 S, 0.5 S and 1 S) — Specification’, IS 15884 ‘Alternating current direct connected static prepayment meters for active energy (Class 1 and Class 2) — Specification’ and IS 15959 ‘Electricity metering — Data exchange for electricity meter reading, tariff and load control — Companion specification’. This standard on the smart meter has been prepared by Bureau of Indian Standards based on the technical specifications and functional requirements published in June 2013 by Central Electricity Authority.’

[Foreword, para 9 (see also Amendment No. 1)] — Substitute the following for the existing:

‘While finalizing this standard a separate standards covering requirements for data exchange specific to smart meter have been prepared, therefore IS 15959 has been revised as follows:

IS 15959 (Part 1) : 2011 Data exchange for electricity meter reading, tariff and load control — Comparison specification: Part 1 Static energy meter (through Amendment)

IS 15959 (Part 2) : 2016 Data exchange for electricity meter reading, tariff and load control — Comparison specification: Part 2 Smart meter

IS 15959 (Part 3) : 2017 Data exchange for electricity meter reading, tariff and load control — Companion specification: Part 3 Smart meter (transformer operated kWh and kVARh, Class 0.2 S, 0.5 S and 1.0 S)’

[Page 1, clause 1.2 b)] — Insert the following at the end:

‘The guidelines for Common Pluggable Communication Module (CPCM) are provided in Annex A — Informative.’

(Page 1, clause 2) — Substitute ‘13779 : 2020’ for ‘13779 : 1999’ and ‘15884 : 2024 for 15884 : 2010’.

(Page 4, clause 6.6, line 3) — Substitute ‘5 years’ for ‘10 years’.

(Page 4, clause 6.8.2, line 2) — Substitute ‘Bureau of Indian Standards Act, 2016’ for ‘Bureau of Indian Standards Act, 1986’.

(Page 5, clause 7.2) — Substitute ‘4.8.2’ for ‘4.6.6.2’.

(Page 6, clause 10.4) — Substitute ‘4.8.2’ for ‘4.6.6.2’.

[Page 4, clause 6.4] — Insert the following at the end:

‘Test shall be applicable for enclosure material used for Common Pluggable Communication Module (CPCM) with temperature applicable for terminal cover and meter case.’

(Page 5, clause 6.10.7) — Delete.

[Page 6, Table 1, Sl No. (iii), col (2)] — Delete ‘Test of influence of immunity to earth fault’.

[Page 6, Table 1, Sl No. (iii), col (3)] — Delete ‘6.10.7’.

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[Page 6, Table 1] — Insert the below test under SI No. (iii) and Insert test at SI No. (vii).

SI No. (1) vii)	Test (2) Test for Load Switch Requirement	Ref to Clause of this Standard (3) 7
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(Page 7) — Insert the following after **11**:

‘ANNEX A

(Informative)

(Clause 1.2)

Guidelines for Common Pluggable Communication Module (CPCM)

A-1 Introduction

The new smart meters may use different types of communication technologies (Cellular/RF Mesh), thus, to enable different communication modules to be used in the same meter, it is necessary to use a universal interface and a specific size irrespective of the choice of communication technology that defines the electrical and mechanical interfaces.

This annexure defines the recommendation for hardware, mechanical features and physical security aspects to ensure that a communication module may be mechanically mounted on different makes of new smart energy meters meeting compliances with IS 16444 (Part 1).

NOTE — Since this new requirement may lead to redesign of meters and certification etc, therefore adherence to this annex can be insisted only after 24 months of issue of this annexure.

A-2 Scope

The annexure covers following key interfaces and details:

- Communication module/hub equipped with RF mesh and cellular communication technology with smart energy meter;
- Mechanical interface;
- Hardware interface
 - Pin definition and details
 - Power supply provided from an energy meter to a communications module;
- Marking on communication module;
- Testing requirements; and
- Security – Physical security.

A-3 Exclusions

The items below are excluded from the scope of this annex:

- PLC/DLC communication module interfaces;
- External antenna provisioning;
- Aesthetics of the communication module and energy meter. It also does not cover the human interface with communication module for (a) General and (b) in field while replacement happens;
- Business Process Flow Details - The document does not cover the module change in field, firmware upgrade, module configuration, data security and other similar business processes/flows from end-to-end business operation perspective which are critical for interoperability implementation and need to be deliberated further during detailed standard and/or guideline document preparation;

- e) Sealing Mechanism - The document does not cover the sealing mechanism/design of communication module on meter body;
- f) Obligation of smart meters' field performance to be agreed between communication module supplier and AMISP supplier/utility after replacement of communication module in field; and
- g) Smart meters having Pluggable Communication Module (PCM).

A-4 Mechanical Interface

A-4.1 General Requirements for Mechanical Interface

Communication module will slide through a pocket given at top hat of meter so that module is accessible from the front side to remove and plug-in with same or different piece of communication module complying with technical specification stated in this annexure. Major components of communication module are:

- a) Module top;
- b) PCB;
- c) Male 12 pin connector; and
- d) Module bottom.

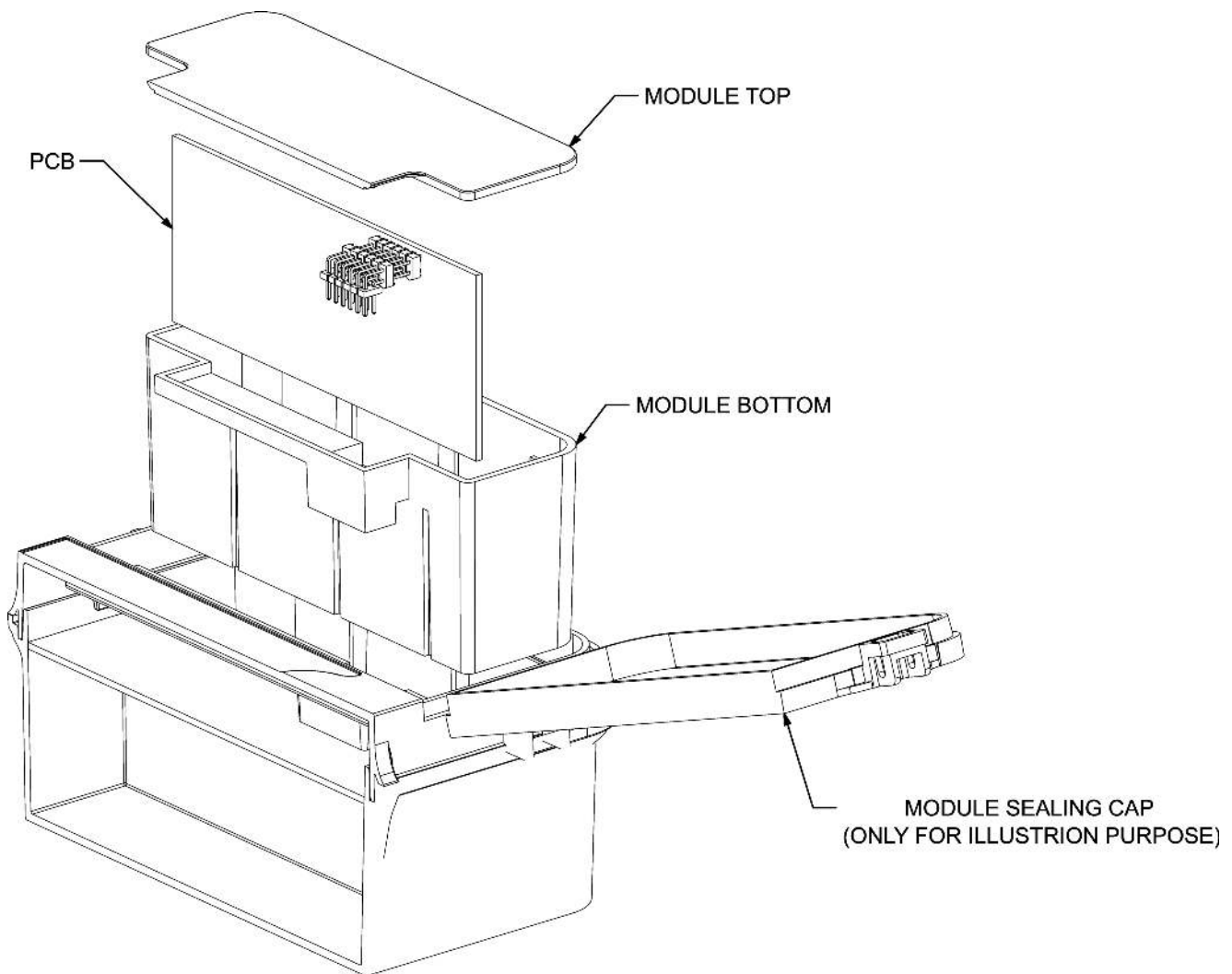


FIG. 1 MODULE ASSEMBLY (FOR ILLUSTRATIVE PURPOSE)

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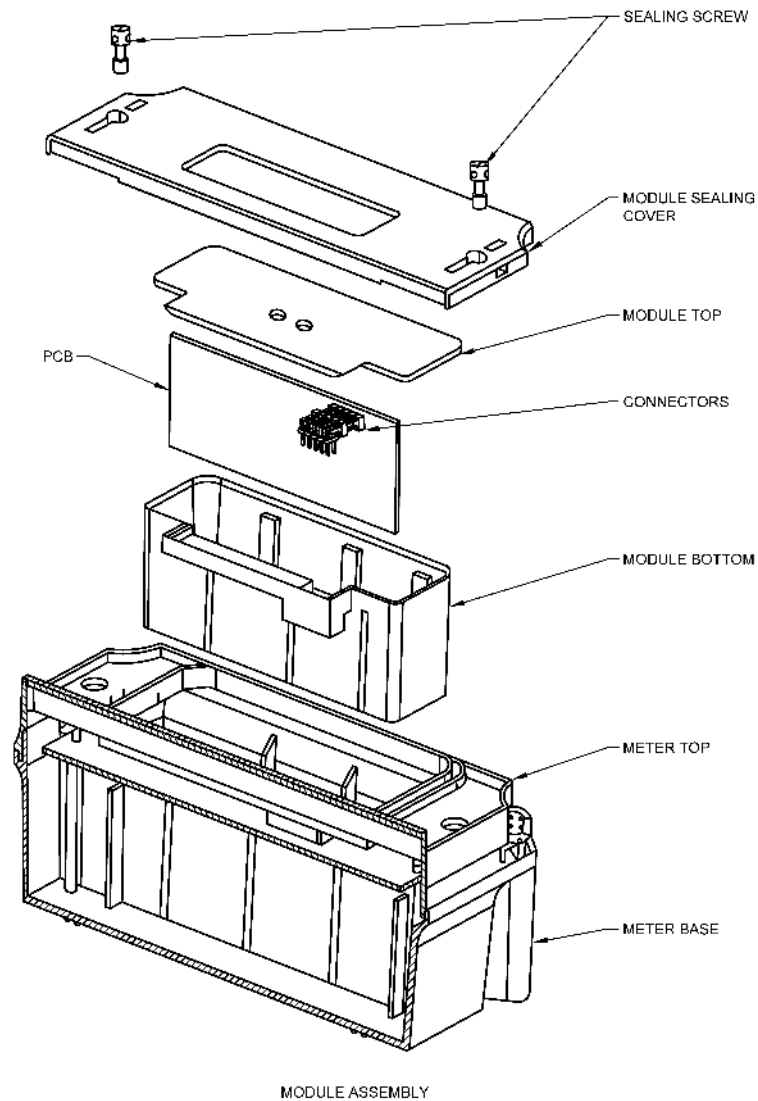


FIG. 2 MODULE ASSEMBLY (FOR ILLUSTRATIVE PURPOSE) - ALTERNATE APPROACH

NOTES

- 1** Orientation of PCB and type of connector shown in above figure is only for illustration purpose. Other design option such as horizontal PCB with straight connector can also be implemented.
- 2** Module sealing cover is the part of meter case. Module sealing cover shown in above figure is only for illustration purpose. Other design options for module sealing cover can be agreed between buyer and seller.

A-4.2 Dimensions

The proposed dimensions of common pluggable module pocket are as defined in the Table 2 below:

Table 2 Proposed Mechanical Dimensions

(Clause A-4.2)

Dimensions of Slot for Communication Module	
L (mm)	115
B (mm)	50
H (mm)	50

NOTE — Refer drawing given in Fig. 3 for tolerances.

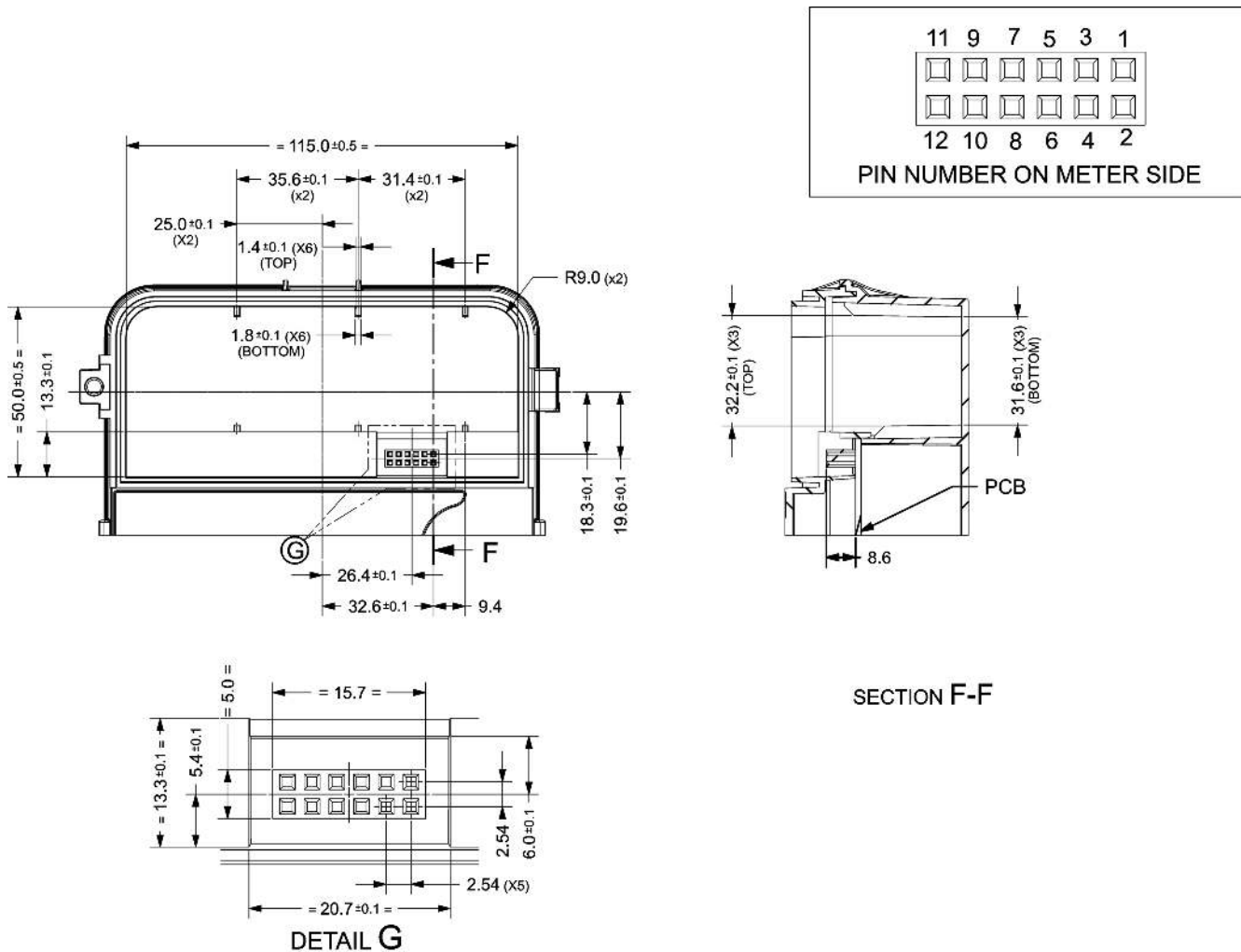


FIG. 3 DIMENSIONAL DRAWING OF MODULE POCKET/SLOT IN METER

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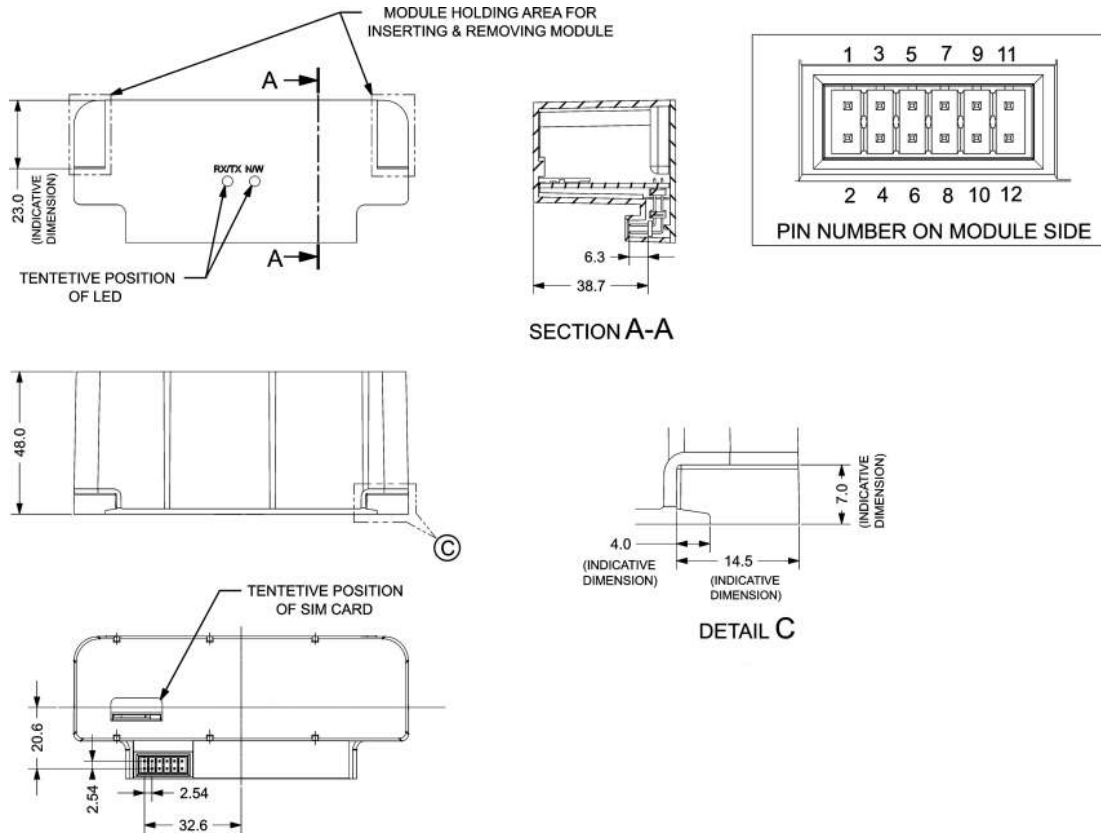


FIG. 4 COMMON PLUGGABLE MODULE CASING DIMENSIONS AND CONNECTOR DETAILS

NOTE — Tentative slot position (for physical SIM/SIMs only), tentative two LED position, module holding slot for removal/insertion (indicative).

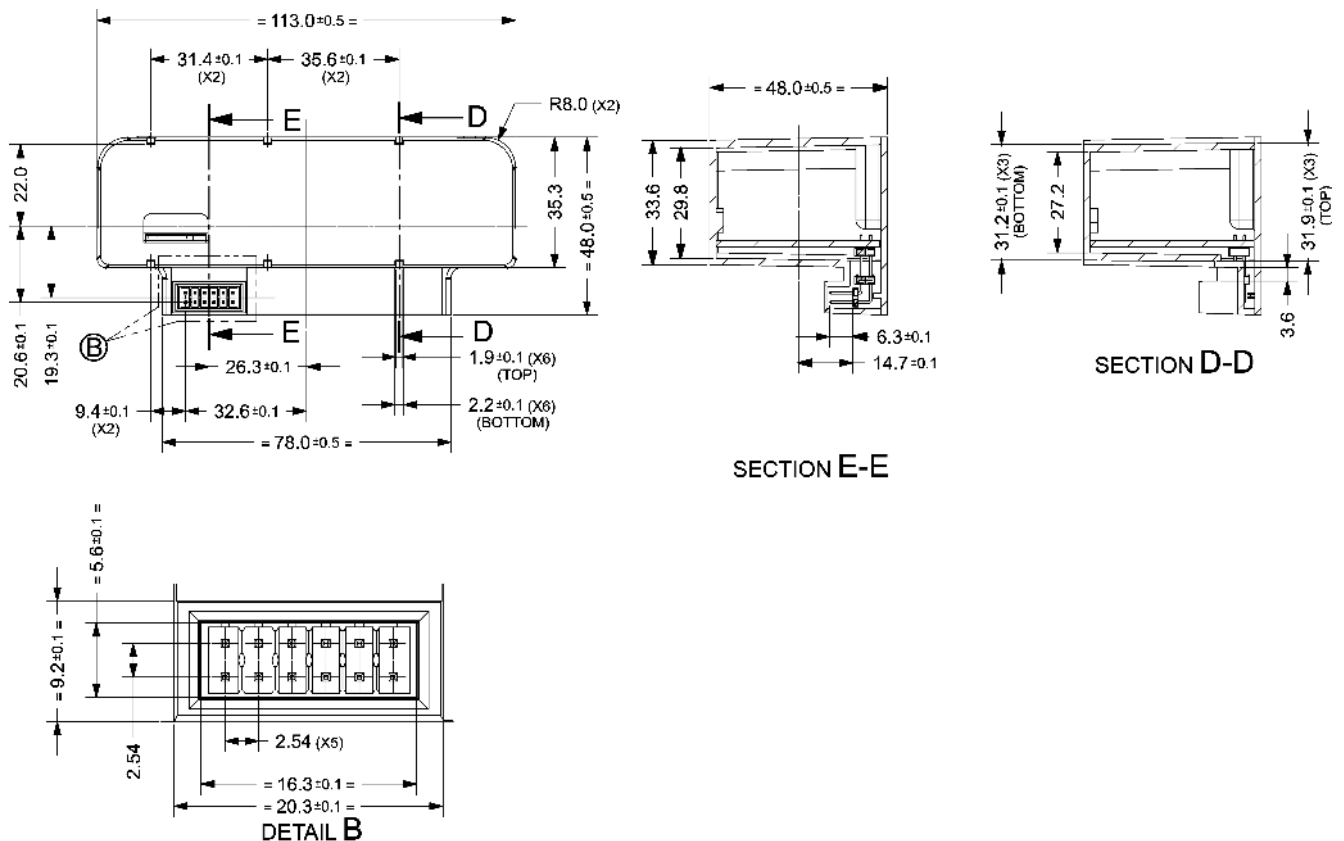


FIG. 5 DETAILS OF CONNECTOR LOCATIONS AND OTHER SIZE PROVISIONS IN MODULE MECHANICALS

NOTE — Length of module mentioned in above drawing shall be treated as maximum. It is possible to design a module which is smaller in length keeping position of connector at same coordinates and module is engaged with maximum possible ribs given in module pocket as defined in Fig. 3.

A-5 Pin Configuration AND Power Supply

A-5.1 Pin Configuration Requirements

The pin configuration of the connector is standardized as given in Table 3 .

Table 3 Pin Configuration

(Clause A-5.1)

NIC Side (Male connector)	
Pin Number	Details
1	GND
2	V _{cc}
3	GND
4	V _{cc}
5	TxD
6	RxD
7	Alert
8	Module Presence Detection
9	Power Fail
10	Reserved
11	GND
12	Reserved

A-5.2 Power Supply Requirements

Communication module uses DC voltage provided from the meter power supply for functioning. V_{cc} is available at Pin 2 and 4 whereas GND is available at Pin 1, 3 and 11. Meter power supply shall provide the DC voltage and current as per the levels defined in Table 4.

Table 4 Power Supply Specifications

(Clause A-5.2)

Power Supply Requirements	
Voltage levels	3.8 V to 4.3V
Current levels	600 mA continuous (Peak to be handled at communication level considering the voltage levels within the defined values as above)
Last gasp super capacitor in module (Yes/No)	Yes
Last gasp super capacitor capacitance (in F, Voltage, Charging Current) - for power capacity	10 F, 2.7 V (min)

NOTE — Above current levels are defined considering LTE Cat 1, so any communication technology having current consumption requirements equal or less than 600 mA can be considered.

The communication module will have a super capacitor to provide power supply to the module in the event of power supply failure to transmit last gasp message. The rating of super capacitor given in Table 3. The power supply necessary for charging the super capacitor derived from the meter supply.

A-5.3 Power Supply Safety Consideration

The power supply provided from the meter may not be isolated. Sufficient care should be taken in module design to prevent any impact. The module design shall ensure that excess current is not drawn beyond the specification defined in this document in normal operating condition as well as in the event of any failure.

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A-5.4 Digital Signalling Pins:

A-5.4.1 Purpose

The connector pins 5 to 9 defined for signalling and used for interaction between meter and the communication modules. The nomenclature for these pins is provided in Table 3. Pin 10 and 12 are reserved currently and it can be used for any additional use cases in future or as per the agreement between meter manufacturer and communication module supplier.

A-5.4.2 Signalling Pin Details

The details of signalling pins described in Table 5.

- a) Data transmission is achieved through TxD and RxD pins;
- b) Alert Pin gives signal to module in case of any event for push notification;
- c) Module presence detection pin gives input to meter on whether the module is present or removed from the meter; and
- d) Power fail pin is an output from the meter, which helps the module to detect power fail event to transmit last gasp message.

Table 5 Data Signaling Pin Details

(Clause A-5.4.2)

Data Signalling Pin Requirements	
Pin	Pin Definition
TxD	Frame format 8, N, 1 (Logic voltage levels to be defined)
RxD	Frame format 8, N, 1 (Logic voltage levels to be defined)
Alert	Pin High- No Alert, Pin Low- Alert received
Module Presence Detection	Pin High- Module removed, Pin Low- Module Present
Power Fail	Pin High- Healthy supply, Pin Low- No supply

NOTE — Pin High: ≥ 2.3 V to 3.4 V, Pin Low: ≤ 0.8 V.

A-6 Marking Details of Communication Module

The module should have following marking details:

- a) Name of manufacturer;
- b) Communication technology, frequency, IMEI number (in case of WAN)/MAC address (in case of RF);
- c) Module type; and
- d) In addition to this, any other information as agreed between buyer and seller.

NOTE — This information shall be readable without opening seals of module sealing cap mounted on meter.

A-7 Other Requirements

CPCM shall be hot swappable with another CPCM.

A-8 Testing Requirements

Case-I: CPCM developed along with Smart Meter by OEM

Smart Meter, with CPCM from OEM or from different manufacturer for obtaining first time license.

The tests specified in IS 16444 (Part 1) shall be conducted by the meter manufacturer for newly developed smart meters equipped with a Common Pluggable Communication Module (CPCM). A BIS license shall be obtained and maintained for the meter and module combination.

Case-II: Smart Meter and CPCM complied with Case-I, and subsequently replaced with different CPCM type (which was part of any approved smart meter type) at manufacturing stage.

The concerned entity shall carry out the schedule of tests as specified in Table 6 below.

In case of RF Technology, CPCM shall comply with the requirement of **9.2** of IS 16444 (Part 1) : 2015.

Table 6 Tests

(Clause A-8)

Sl No.	Tests to be Conducted
(1)	(2)
i)	Power consumption as per 6.10.1 of IS 16444 (Part 1) : 2015
ii)	Test for electromagnetic compatibility as per 6.11 of IS 16444 (Part 1) : 2015
iii)	Test of limits of error due to variation of the current as per 11.1 of IS 13779 : 2020
iv)	Insulation requirements as per 6.10.6 of IS 16444 (Part 1) : 2015
v)	Influence of supply voltage as per 6.10.2 of IS 16444 (Part 1) : 2015
vi)	Mechanical requirements as per 6.5 of IS 16444 (Part 1) : 2015
vii)	Tests listed in Note 1 (b) and 1 (c) of Table 1 of IS 16444 (Part 1) : 2015

Case-III: Smart Meter and CPCM complied with Case-I, and subsequently CPCM is replaced with different module type (which is not part of any approved smart meter type) at manufacturing stage.

The tests specified in Table 7 below shall be conducted by the concerned entity.

In case of RF Technology, CPCM shall comply with the requirement of **9.2** of IS 16444 (Part 1) : 2015.

Table 7

(Clause A-8)

Sl No.	Tests to be Conducted
(1)	(2)
i)	Power consumption as per 6.10.1 of IS 16444 (Part 1) : 2015
ii)	Test for electromagnetic compatibility as per 6.11 of IS 16444 (Part 1) : 2015
iii)	Test of limits of error due to variation of the current as per 11.1 of IS 13779 : 2020
iv)	First breath (power on) and last gasp (power off) as per 11.7 of IS 16444 (Part 1) : 2015
v)	Insulation requirements as per 6.10.6 of IS 16444 (Part 1) : 2015
vi)	Influence of supply voltage as per 6.10.2 of IS 16444 (Part 1) : 2015
vii)	Mechanical requirements as per 6.5 of IS 16444 (Part 1) : 2015
viii)	Climatic condition as per 6.9 of IS 16444 (Part 1) : 2015
ix)	Tests listed in Note 1 (b) and 1 (c) of Table 1 of IS 16444 (Part 1) : 2015

Case-IV: Smart Meter and CPCM complied with Case-I under service condition, and subsequently CPCM is replaced with different module type (which was part of any approved smart meter type)

The concerned entity shall carry out the schedule of tests as specified in Table 6 on CPCM with same type of meter in which the CPCM has to be replaced.

NOTE — Due process of replacement of different modules shall be done after the inclusion of the variety in the BIS licence of the CPCM module.

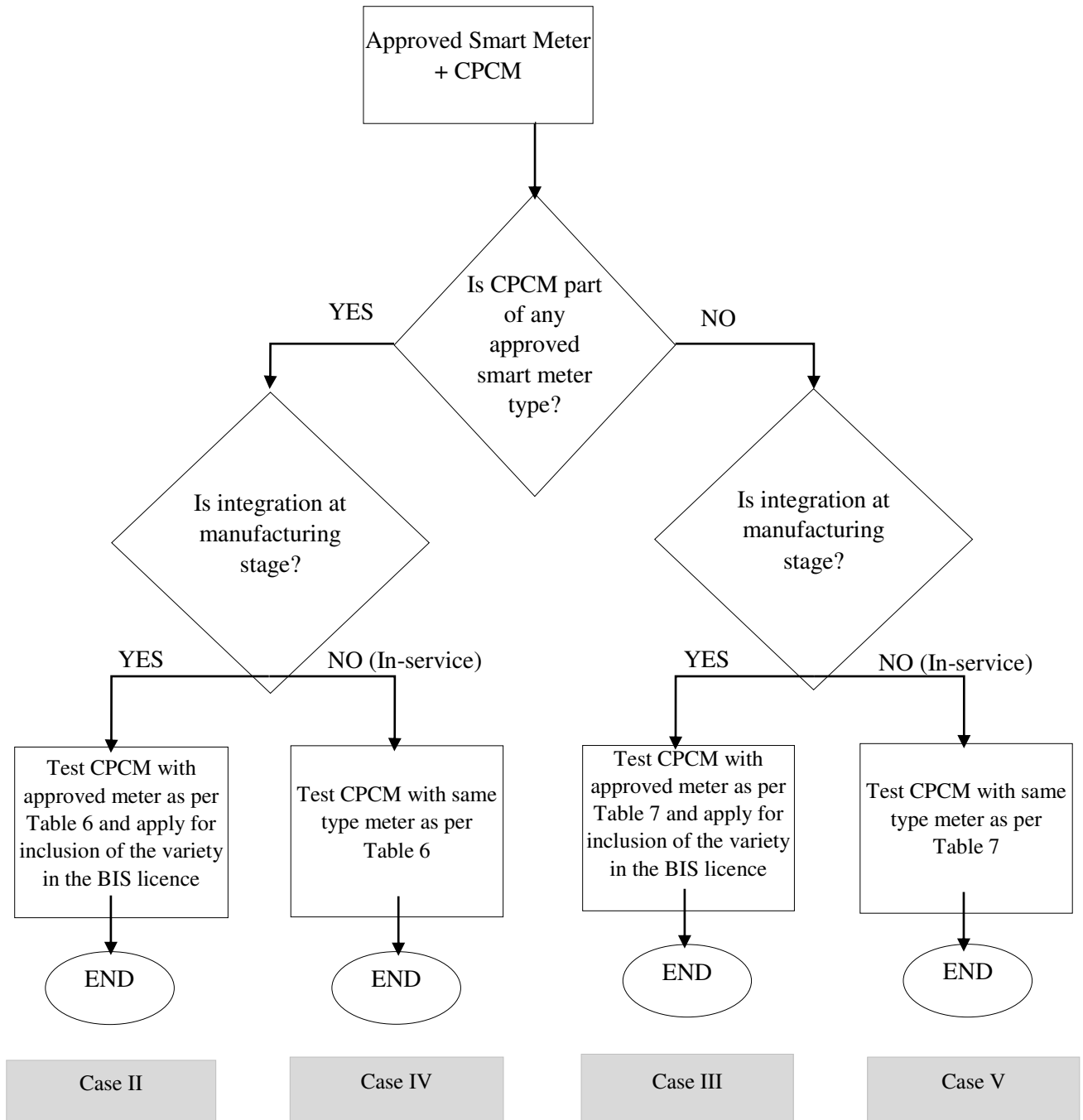
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Case-V: Smart Meter and CPCM complied with Case-I under service condition, and subsequently CPCM is replaced with different module type (which was not part of any approved smart meter type)

The concerned entity shall carry out the schedule of tests as specified in Table 7 on CPCM with same type of meter in which the CPCM has to be replaced.

NOTE — Due process of replacement of different modules shall be done after the inclusion of the variety in the BIS licence of the CPCM module.

The testing methodology on CPCM (Case II to V) have been illustrated in Fig 6.



NOTE —

- (a) CPCM: Common Pluggable Communication Module.
- (b) In service: Approved smart meter in installed condition at consumer.

FIG. 6 TESTING METHODOLOGY ON CPCM WITH APPROVED SMART METER (CASE II TO V)

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A-9 MTCTE Certification

MTCTE certification for the common pluggable communication module (CPCM) shall be obtained in accordance with the latest essential requirements issued by the Telecommunication Engineering Centre (TEC), Government of India.

A-10 Security

A-10.1 Physical Security

Mechanical security: Mechanical security for the communication module is to be ensured by due sealing mechanism as agreed between buyer and seller.

A-10.2 Tamper Detection

Communication module removal detection electronically is to be ensured by pins described in signalling pin details section above.

A-10.3 Data Security

Managing data communication security algorithms across smart metering eco system including data security is per applicable standard and cyber security guideline

A-11 Definitions/Glossary

Table 8 below provides information of key terms and abbreviations used in the annexure for better understanding and clarity.

Table 8 Definition/Glossary

(Clause A-11)

Sl No. (1)	Term (2)	Definition or Explanation (3)
i)	AMISP	Advanced Metering Infrastructure Service Provider
ii)	CPCM	Common Pluggable Communication module for Smart Meter
iii)	MTCTE	Mandatory Testing and Calibration of Telecommunication Equipment
iv)	OEM	Original Equipment Manufacturer
v)	PCB	Printed Circuit Board
vi)	TEC	Telecommunication Engineering Centre - Government of India